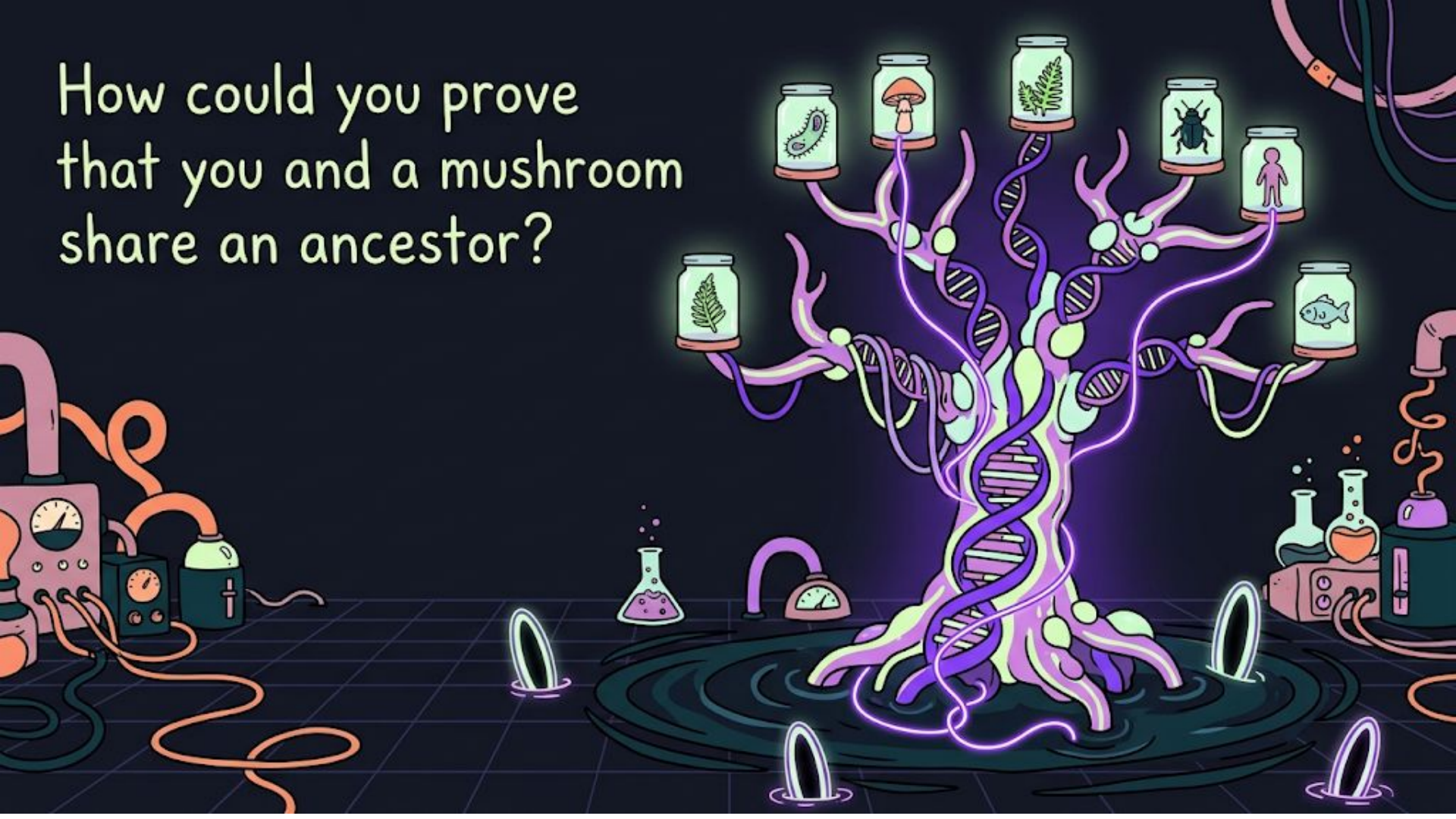
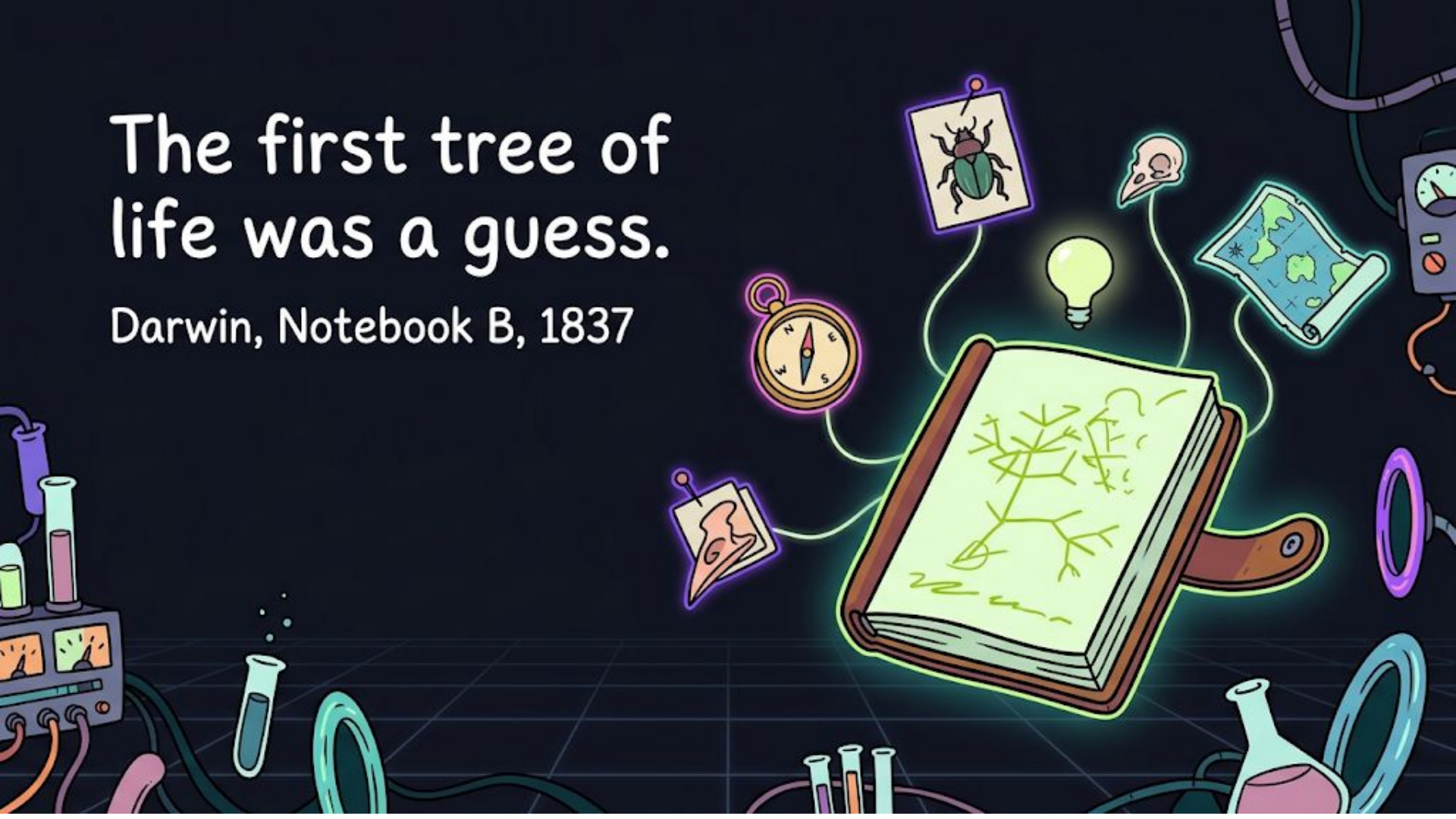
How could you prove that you and a mushroom share an ancestor?



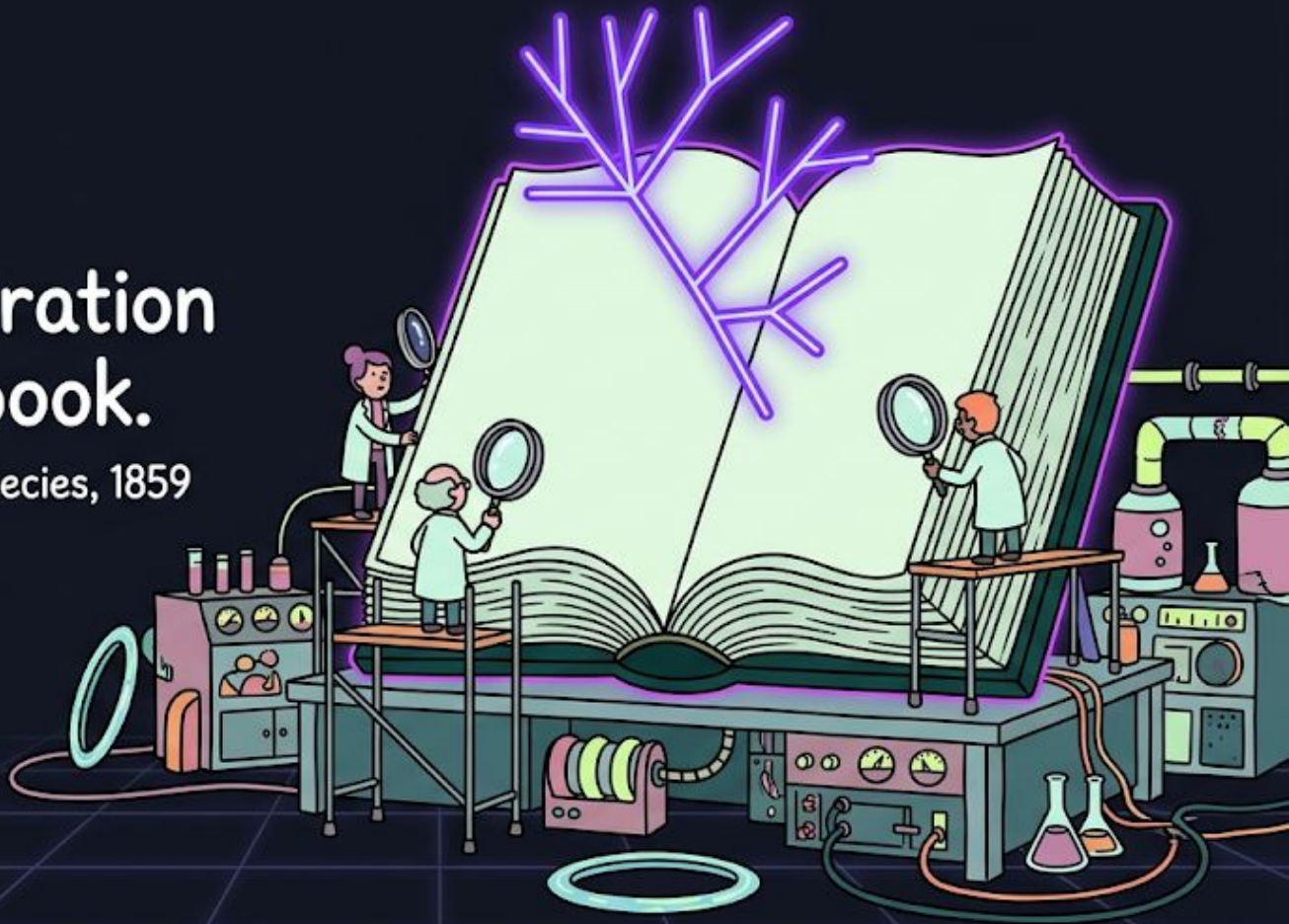
The first tree of life was a guess.

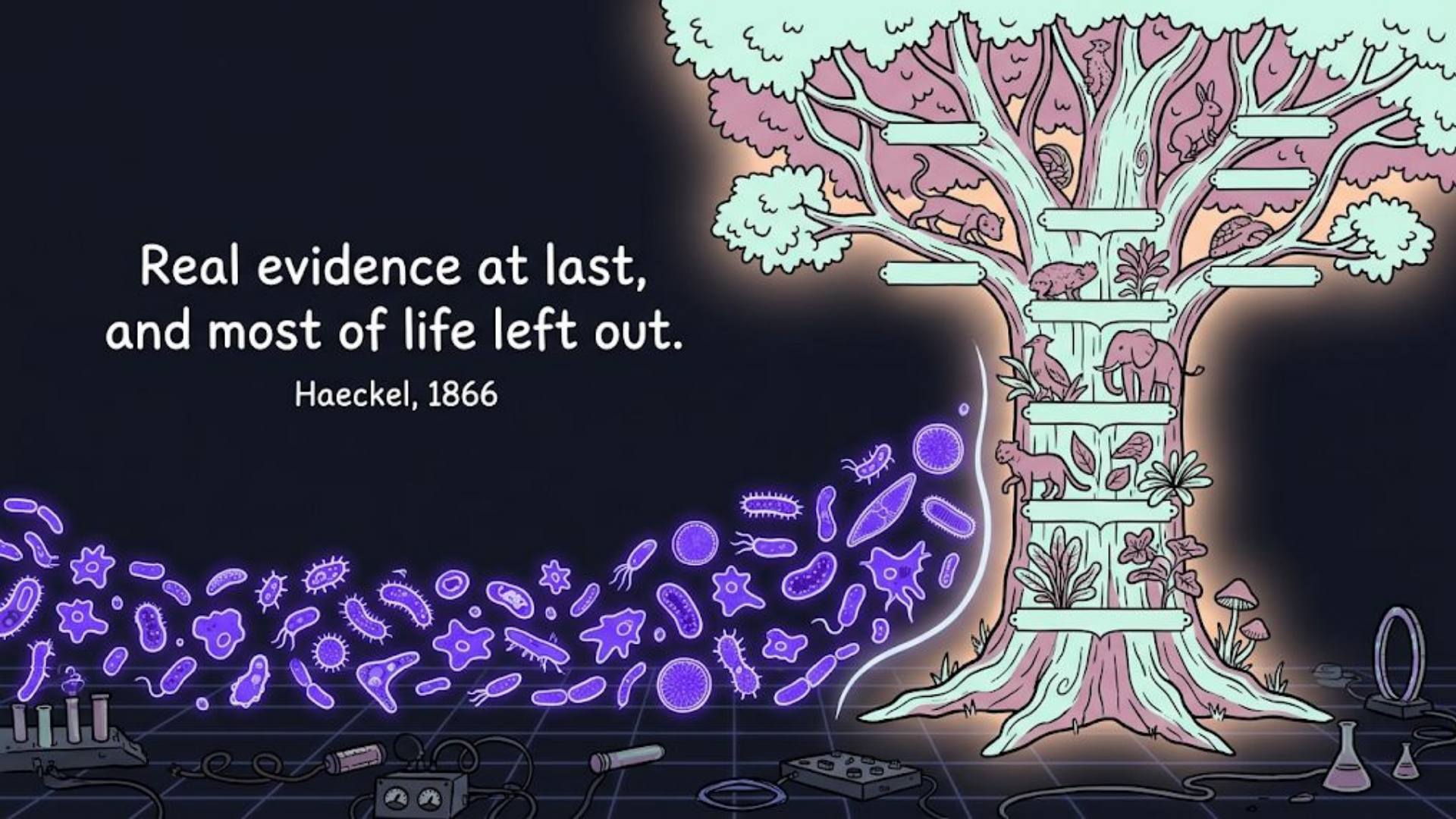
Darwin, Notebook B, 1837



The only illustration in the whole book.

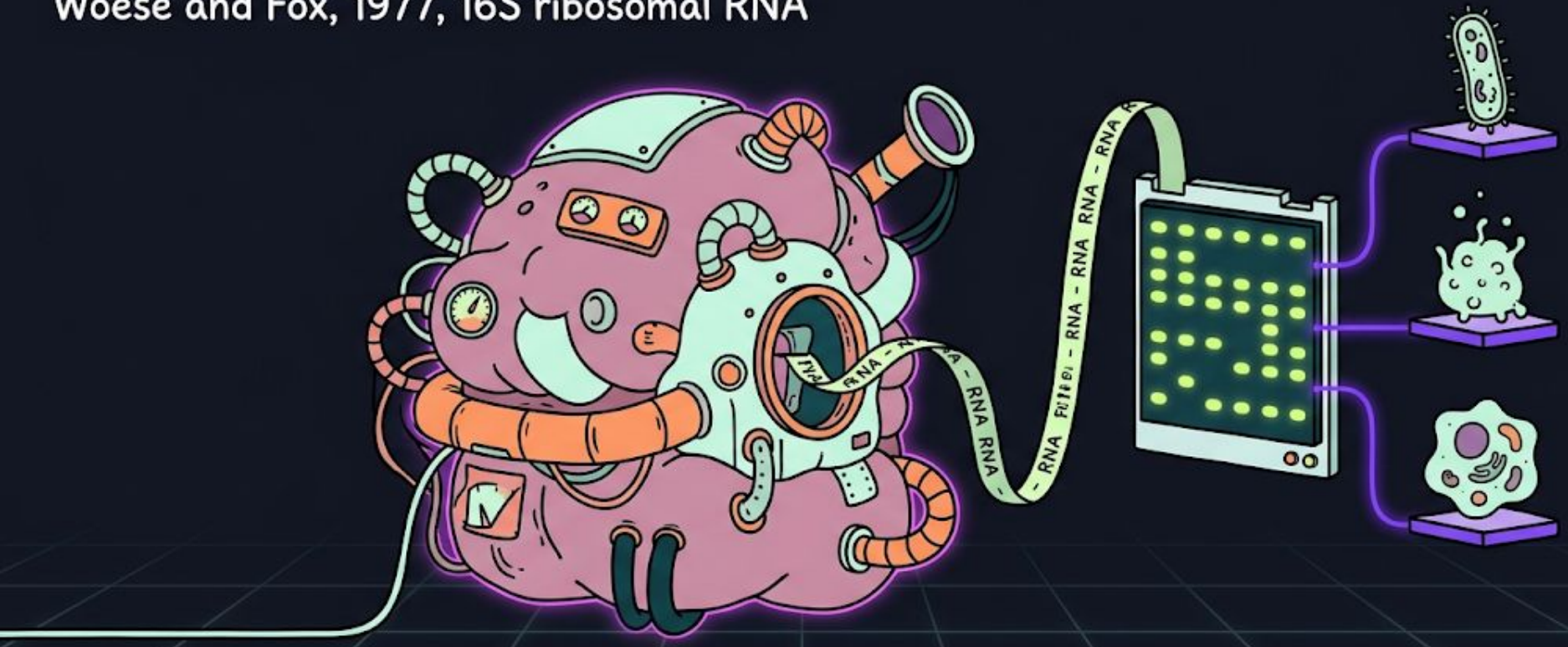
Darwin, On the Origin of Species, 1859




Real evidence at last,
and most of life left out.
Haeckel, 1866

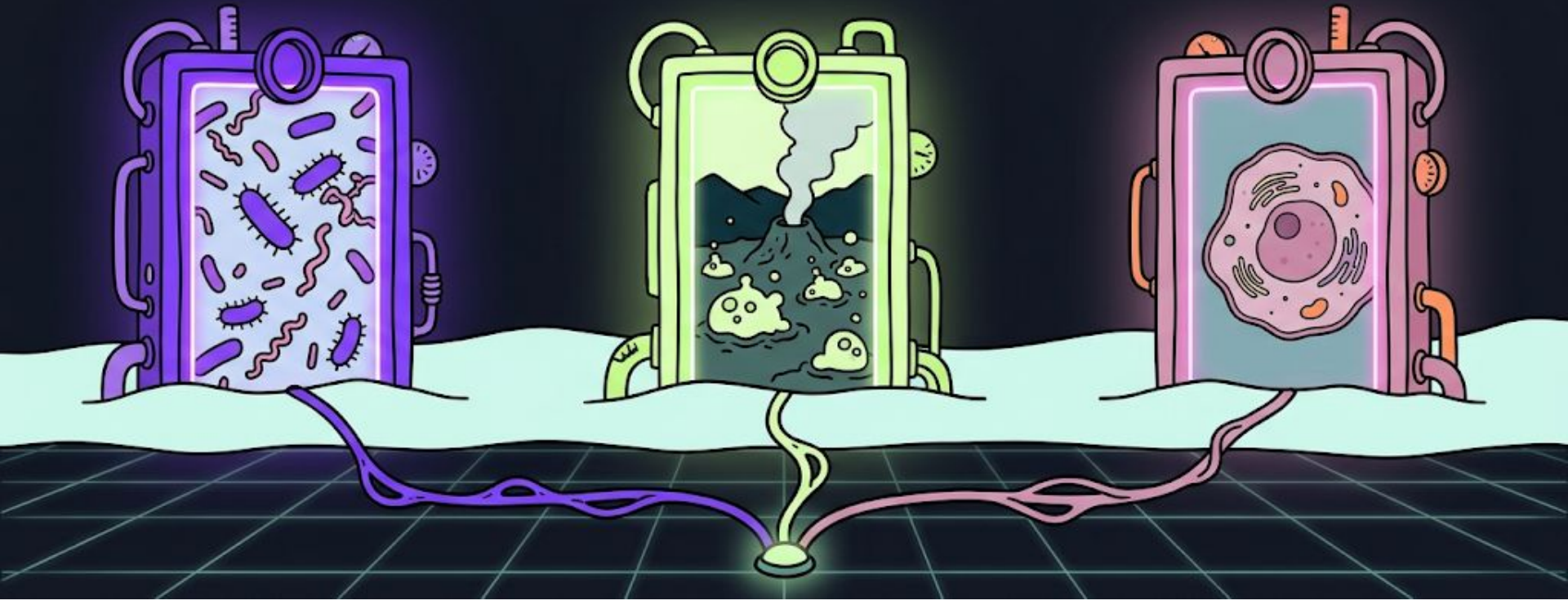
Stop looking at bodies. Read one molecule.

Woese and Fox, 1977, 16S ribosomal RNA



Three domains, finally named.

Woese, Kandler and Wheelis, 1990



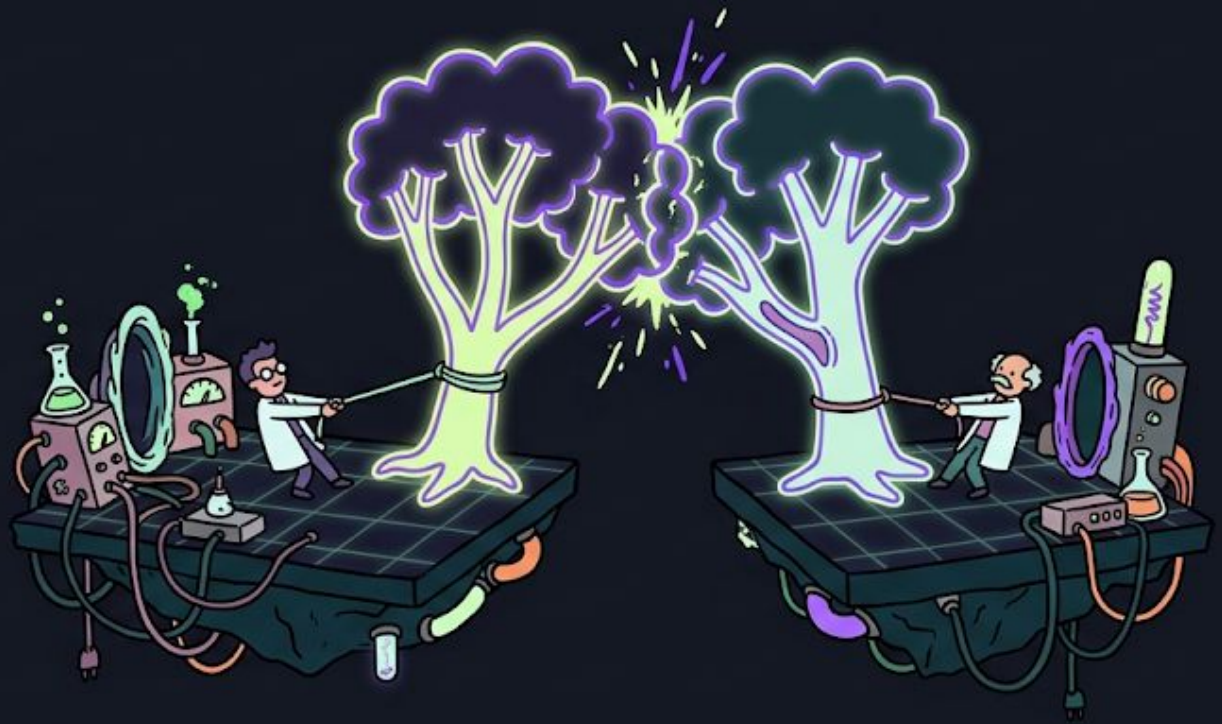
Animals are a twig.

Hug et al., 2016, 3,083 genomes



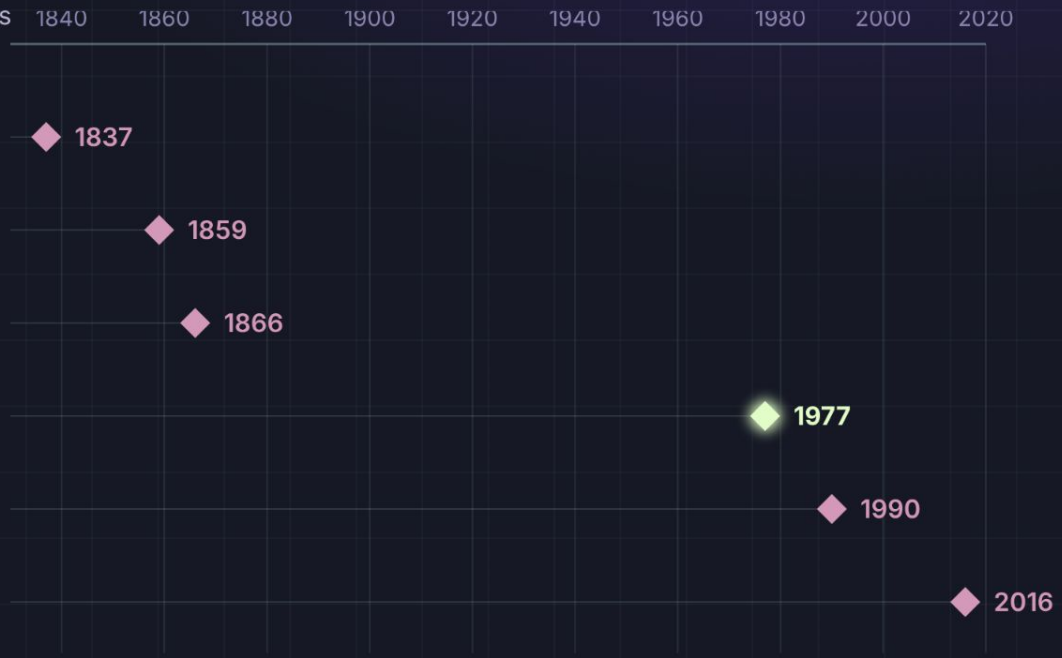
Two domains or three? Still open.

the argument is live right now



Every new kind of evidence redrew the tree

From a sketch with no data to 3,083 genomes



111 years between the anatomical tree and the molecular one — they were waiting for a method

PART 1 · BUILD A TREE · 5 MINUTES

You have four sequences. Draw the tree.

Twenty bases each. No software, no formula — just count.

Seq 1	AAGCTTCACGTTGCAACGTA
Seq 2	AAGCTTCACGTTGCAACGTG
Seq 3	AAACTTCATGTTGGAATGTA
Seq 4	TCGCACCACGACGCTACGCA

	Seq 1	Seq 2	Seq 3	Seq 4
Seq 1	—			
Seq 2		—		
Seq 3			—	
Seq 4				—

1. Count the differences between every pair — fill the matrix.
2. Join the closest pair first.
3. Add the next closest, then the next.
4. Use **Seq 4** to root the tree.

PART 1 · BUILD A TREE · KEY

Sequences 1 and 2 pair first; 4 sets the root

Highlighted bases differ from Seq 1. Seq 4 is the outgroup — it gives the tree its direction.

Seq 1	AAGCTTCACGTTGCAACGTA
Seq 2	AAGCTTCACGTTGCAACGT**G**
Seq 3	AA**A**CTTCA**T**GTTG**G**AA**T**GTA
Seq 4	**T****C**GC**A****C**CACG**A****C**GC**T**ACG**C**A

	Seq 1	Seq 2	Seq 3	Seq 4
Seq 1	—	1	4	8
Seq 2	1	—	5	9
Seq 3	4	5	—	12
Seq 4	8	9	12	—

JOIN THE CLOSEST PAIR FIRST



THE TREE

((1, 2), 3), 4

Every unit hangs on a branch.

the year map, part 2



Viruses are not on the tree.

no ribosomes, nothing to compare





Life Sciences · ages 15–18 · 2026–27

Year map · 2 h/week ≈ 60 h left: 56 h new + 4 h review · on-site S2–S4 in Hochrindl · aligned to IB DP & AP

Biomolecules & enzymes

IB DP A1.1, B1.1–B1.2, C1.1 · AP Bio U1, U3 · 5 h

- ○ Water & hydrogen bonding
- ○ Carbohydrates, lipids & proteins
- ○ Protein structure: four levels
- ○ Enzymes: active site, denaturation, inhibition
- ○ Condensation & hydrolysis

On-site: catalase vs temperature & pH · food tests

Cells & membranes

IB DP A2.2, B2.1–B2.3 · AP Bio U2 · 5 h

- ○ Prokaryotic vs eukaryotic cells; endosymbiosis
- ○ Membranes: fluid mosaic model
- ○ Diffusion, osmosis, active transport
- ○ Surface area to volume ratio

On-site: agar-cube diffusion · potato osmosis series

Molecular genetics & biotechnology

IB DP A1.2, D1.1–D1.3, D2.2 · AP Bio U6 · 4 h

- ✓ **Genes; DNA vs RNA; nucleotides & nucleosides**
- ✓ **Transcription & translation: gene → protein**
- ✓ **Chromosomes & chromatin**
- ✓ **Genetic & epigenetic control of replication & repair**
- ✓ **Mutations: neutral, beneficial, harmful**
- ✓ **IVF, cloning & gene editing**
- ○ Gene regulation: operons & transcription factors
- ○ PCR, gel electrophoresis & sequencing

On-site: food-dye gel electrophoresis

Cell division & inheritance

IB DP D2.1, D3.1–D3.2 · AP Bio U4–U5 · 6 h

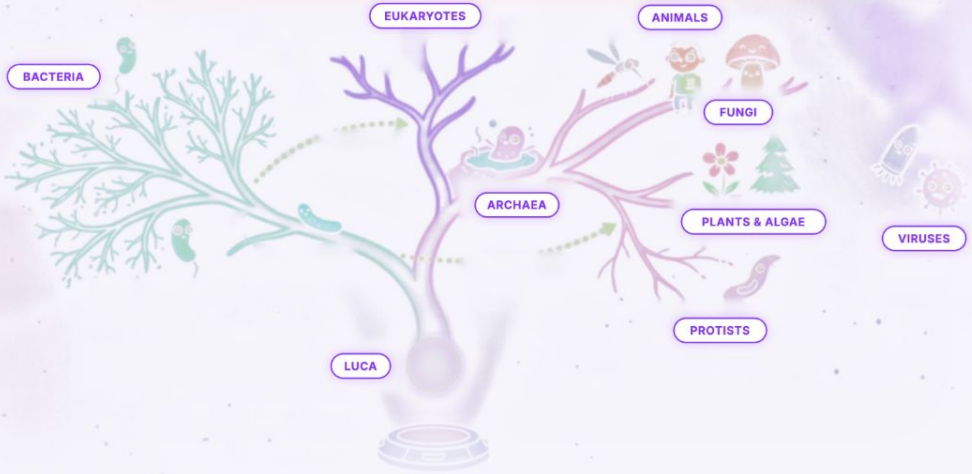
- ✓ **Somatic vs germ cells**
- ○ Mitosis, cell-cycle control & cancer
- ○ Meiosis & sources of variation
- ○ Mendelian & non-Mendelian inheritance; pedigrees
- ○ Chi-square test

On-site: onion root-tip mitosis

Thinking like a scientist

IB DP Nature of science, Tools & Inquiry · AP science practices · in every topic

- ✓ **Causality: Bradford Hill's guidelines** · ○ Research question, variables & controls
- ○ Statistics: error bars, chi-square, t-test · ○ Evaluating study designs
- ○ Lab reports & IB internal assessment



Atoms, bonding & the mole

IB DP S1.2–S1.4, S2.1–S2.3, S3.1 · AP Chem U1–U3 · 10 h

- ○ Atomic structure, isotopes, electron configuration
- ○ Periodic trends
- ○ Ionic, covalent & metallic bonding
- ○ Shape, polarity & intermolecular forces
- ○ The mole, concentration & stoichiometry

On-site: polarity & solubility tests · molecular model kits

Reactions & energy

IB DP R1.1, R2.2–R2.3, R3.1–R3.2 · AP Chem U4–U8 · 7 h

- ○ Redox & reaction types
- ○ Enthalpy: exo- & endothermic reactions
- ○ Rates, activation energy & catalysts
- ○ Equilibrium & Le Chatelier's principle
- ○ Acids, bases, pH & buffers

On-site: tablet fizz rates vs temperature · vinegar titration · buffer test

Energy in cells

IB DP C1.2–C1.3 · AP Bio U3 · 5 h

- ○ ATP: the energy currency
- ○ Respiration: glycolysis, Krebs, electron transport
- ○ Photosynthesis: light reactions & Calvin cycle
- ○ Fermentation

On-site: floating leaf discs · yeast CO_2 vs temperature

Evolution & ecology

IB DP A3.2, A4.1–A4.2, C4.1–C4.2, D4.1–D4.3 · AP Bio U7–U8 · 8 h

- ✓ **Natural & artificial selection for the market**
- ✓ **Co-evolution: sickle cell, Plasmodium, Anopheles**
- ○ Hardy–Weinberg & speciation
- ○ Phylogenetics & cladograms
- ○ Energy flow & carbon cycle
- ○ Populations & biodiversity indices
- ○ Climate change

On-site: alpine quadrats & Simpson's index · stream biotic index

Microbes, immunity & body systems

IB DP B3.1–B3.2, C2.2, C3.1–C3.2, D3.3 · AP Bio U4 · 6 h

- ✓ **Environment changes gene expression: Salmonella in space**
- ✓ **More aggressive biofilms on spacecraft**
- ○ Viruses & bacteria; antibiotic resistance
- ○ Immune response & vaccines
- ○ Nerves & hormones
- ○ Homeostasis: blood glucose & temperature
- ○ Gas exchange & circulation

On-site: heart & breathing rate at altitude · reaction time

✓ **covered at on-site session 1**
○ planned by end of year
On-site: experiment for sessions 2–4 in Hochrindl
Grey line: IB DP topic codes · AP course units · hours